Game Theoretic Modeling of Jamming Attack in Wireless Powered Networks

Dusit Niyato[1], Ping Wang[1], Dong In Kim[2], Zhu Han[3], and Lu Xiao[1]
[1] School of Computer Engineering, Nanyang Technological University (NTU), Singapore
[2] School of Information and Communication Engineering, Sungkyunkwan University (SKKU), Korea
[3] Department of Electrical and Computer Engineering, University of Houston, Texas, USA

***Abstract*—In wireless powered networks, a user can make a request and use the wireless energy transferred from an energy source for its data transmission. However, due to broadcast nature of wireless energy transfer (e.g., RF energy), a malicious node (i.e., an attacker) can also intercept the energy and use it to perform an attack by jamming the data transmission of the user. We consider such a jamming attack where the user and attacker are aware of each other. We formulate a game theoretic model to analyze the energy request and data transmission policy of the user and the attack policy of the attacker when the user and the attacker both want to maximize their own rewards. We use an iterative algorithm designed based on the best response dynamics to obtain the solution defined in terms of the constrained Nash equilibrium. The numerical results show not only the convergence of the proposed algorithm, but also the optimal reward of the user under different energy cost constraints.**

***Index Terms*—Wireless powered networks, wireless jamming, constrained stochastic game.**

I. INTRODUCTION

Wireless powered networks utilize different techniques (e.g., RF, inductive coupling, and magnetic resonance coupling) to transfer wireless energy from a source to a user's device [1]. The user uses that energy for data transmission. However, in the network, a malicious node (i.e., an attacker) may exist to perform jamming attack, disrupting data transmission of the user [2]. The attacker may receive and utilize the energy transferred from the source for the attack. Hence, the network faces the problem of energy transfer as it not only incurs higher cost, but also facilitates the attacker to perform more attacks. The user has to optimize his/her data transmission strategy and also the wireless energy transfer policy. Likewise, given the available wireless energy, the attacker can optimize its attack strategy.

In wireless energy harvesting and wireless powered networks, most of the work related to security is to analyze and optimize secrecy rate [3], [4], [5], [6], [7], [8]. For example, the authors of [3] optimized beamforming of the nodes in the network with passive eavesdroppers. The objective is to minimize transmit power given that the receiver can harvest RF energy from the transmission. The authors of [7] introduced secure relay beamforming (SRB) scheme for simultaneous wireless information and power transfer (SWIPT). The scheme is to maximize secrecy rate under harvested energy and transmit power of the relay. However, none of the work considered a jamming attack in wireless energy harvesting/transfer networks. In traditional networks, such a jamming attack issue was analyzed using game theory [9], [10], [11], [12] to find equilibrium strategies of a legitimate user and attacker. Stochastic optimization was also used [13] to find an optimal transmission policy under the attack.

In this paper, we consider a wireless powered network with a legitimate user and attacker. The user requests for wireless energy transfer from an energy source, and uses the energy for data transmission to a receiver. The attacker can also intercept the energy and use it to perform the jamming attack. We formulate a constrained stochastic game model to analyze the energy request and data transmission policy of the user and the attack policy of the attacker. The user aims to maximize the reward in terms of utility from transmitting data minus energy cost. The attacker aims to maximize the successful jamming attack. We consider a constrained Nash equilibrium as a solution. The performance evaluation shows the convergence of the policies to the equilibrium. Furthermore, we show that if the energy budget of the user can be controlled, the user can achieve optimal reward.

II. SYSTEM MODEL

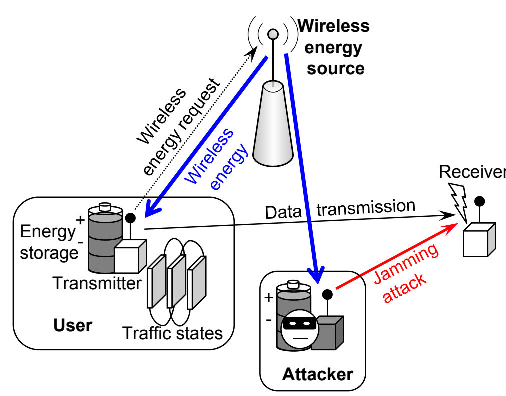


Fig. 1. System model.

We consider a wireless powered network as shown in Fig. 1. The network is composed of a legitimate user who can request for wireless energy transfer (e.g., RF energy) from the wireless energy source. The user's device has an energy storage with a finite capacity (i.e., E_U units of energy). The received energy will be stored in the storage. The amount of wireless energy received by the user's device depends on the channel quality (i.e., a channel state) which is random (e.g., due to fading effect). The set of channel states is denoted by $\mathcal{C}_{\mathrm{U}} = \{0, 1, \dots, C\}$ where C is the highest channel states. Without loss of generality, we assume that at channel state $c \in \mathcal{C}_{\mathrm{U}}$, c units of energy can be received by the user's device if the wireless energy source transfers energy (i.e., the user makes a request). Let $\mathbf{C}_{\mathrm{U}}$ denote the transition probability matrix of the channel state. Its element is denoted by $C^{\mathrm{U}}_{c,c'}$ which is the probability that the current channel state is c and the next channel state is c'. If the user requests for wireless energy, it will incur the cost of μ. This cost can be the reserved energy needed to make a request.

The user generates different types of traffic. The type of traffic depends on the traffic state, whose set is denoted by $\mathcal{D} = \{1, \dots, D\}$ where D is the highest traffic state. Let $\mathbf{D}$ denote the transition probability matrix of the traffic state. Its element is denoted by $D_{d,d'}$ which is the probability that the current traffic state is d and the next traffic state is d'. The user's device requires one unit of energy from the storage to transmit the data to the receiver in any traffic state. We consider a real-time traffic in which if the data is generated (according to the traffic state) but cannot be transmitted (e.g., due to lack of energy), it will be discarded immediately. If the generated data at traffic state $d \in \mathcal{D}$ can be transmitted successfully, the user will gain the utility of u^{U}_d. Otherwise, the user receives zero utility.

In addition to the legitimate user, the network has an attacker that performs a jamming attack to the data transmission by the user. The attacker quietly receives wireless energy when the user makes a request. The amount of energy received by the attacker is random and depends on the channel state. The channel state transition probability matrix is denoted by $\mathbf{C}_{\mathrm{A}}$. The attacker has an energy storage with the capacity of E_{A} units of energy. If the attacker performs the jamming attack, it will consume one unit of energy from the storage. Note that the energy unit of the user and attacker can be different. For example, one unit of energy of the attacker could be larger than that of the user since the attacker requires more energy to successfully jam the data transmission of the user. If the attacker successfully jams the data transmission of the user, it gains the utility of u_{A}. Otherwise, the attacker receives zero utility.

In the aforementioned wireless powered network, the user is aware that there is an attacker. The user has to decide whether to request for wireless energy and transmit the data or not depending on his/her channel state, traffic state, and energy level of his/her own energy storage. Similarly, the attacker has to decide whether to perform the jamming attack or not based on its channel state and energy level of its own energy storage.

In the following, we will develop a game theoretic model for the user and attacker to obtain their equilibrium policies.

III. Game Formulation between User and Attacker

We analyze the energy request and data transmission policy of the user and investigate the jamming attack policy of the attacker by formulating a constrained stochastic game model [14]. The game model is composed of two players, which are the user and attacker in the network. The set of players is denoted by $\mathcal{N}$. Each player is defined by the tuple $\{\Omega_n, \mathcal{A}_n, \mathbf{P}_n, \mathscr{C}_n, \mathcal{V}_n\}$ where $n \in \{\mathrm{U}, \mathrm{A}\}$ for the user and attacker, respectively.

- Ω_n is a finite local state space of player n. The local state is denoted as $\omega_n \in \Omega_n$, and $\Omega = \Omega_{\mathrm{U}} \times \Omega_{\mathrm{A}}$ is a global state space, where $\times$ is the Cartesian product.
- $\mathcal{A}_n$ is a finite local action space. $\mathcal{A} = \mathcal{A}_{\mathrm{U}} \times \mathcal{A}_{\mathrm{A}}$ is a global action space. The global action is defined as $\hat{\mathbf{a}} = (\mathbf{a}_{\mathrm{U}}, a_{\mathrm{A}})$.
- $\mathbf{P}_n(\mathbf{a})$ is a transition probability for player n given action $\mathbf{a}$.
- $\mathscr{R}_n$ is an immediate reward function.
- $\mathcal{V}_n = \{\mathscr{V}_{n,j}\}$ is a set of constraints.

In the following, we present the definition and derivation of the above components of each player. We may omit the indexes $_{\mathrm{U}}$ and $_{\mathrm{A}}$ for the user and attacker when there is no ambiguity from the context.

A. User Side Analysis

The state space of the user is defined as follows:

$$\Omega_{\mathrm{U}} = \Big\{ \big(\mathbb{E}, \mathbb{C}, \mathbb{D}\big); \mathbb{E} \in \{0, 1, \dots, E_{\mathrm{U}}\} \Big\}, \tag{1}$$

where $\mathbb{E}$ is the energy level of the energy storage, $\mathbb{C} \in \mathcal{C}_{\mathrm{U}}$ is the channel state, and $\mathbb{D}$ is the traffic state of the user. The composite state variable of the user is denoted by $\omega_{\mathrm{U}} = (e, c, d)$.

The action space is defined as $\mathcal{A}_{\mathrm{U}} = \{(a_1, a_2)\}$. $a_1 \in \{0, 1\}$ is the decision for requesting and not requesting for wireless energy transfer, respectively. $a_2 \in \{0, 1\}$ is the decision for transmitting and not transmitting data, respectively. $\mathbf{a} \in \mathcal{A}_{\mathrm{U}}$ is a composite action variable.

Next we derive the transition matrix for the user. We first consider the transition of the energy level. Let $\mathbf{E}_\delta$ denote the transition matrix when the net change of the energy level is δ, where $\delta = -1, 0, 1, \dots, C$. The energy level decreases by one unit (i.e., $\delta = -1$) if no energy arrives (e.g., the channel state is $c = 0$ and the user transmits data). The element of matrix $\mathbf{E}_\delta$ is denoted by $E^{\delta}_{e,e'}$ which is the probability that the energy level changes from e to e'. The probability is defined as follows:

$$E^{\delta=-1}_{e,e'} = \begin{cases} 1, & e' = \max(0, e-1), \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

$$E^{\delta=0}_{e,e'} = \begin{cases} 1, & e = e', \\ 0, & \text{otherwise}, \end{cases} \tag{3}$$

$$E^{\delta>0}_{e,e'} = \begin{cases} 1, & e' = \min(E_{\mathrm{U}}, e+\delta), \\ 0, & \text{otherwise}. \end{cases} \tag{4}$$

Let $\mathbf{T}$ denote the transition matrix combining the transitions of the channel state and traffic state. It is obtained from $\mathbf{T} = \mathbf{C}_{\mathrm{U}} \otimes \mathbf{D}$, where $\otimes$ is the Kronecker product. Each element of the matrix $\mathbf{T}$ corresponds to the channel state c and traffic state d. Now, we combine with the transition of the energy level. Let $T_{(c,d),(c',d')}$ denote the element of the matrix $\mathbf{T}$, which is the probability that channel state c and traffic state d change to c' and d', respectively. The transition matrix of the user, denoted by $\mathbf{P}_{\mathrm{U}}(\mathbf{a})$, is expressed as follows.

$$\mathbf{P}_{\mathrm{U}}(\mathbf{a}) = \begin{bmatrix} T_{(0,0),(0,0)}\mathbf{E}_{\delta(\mathbf{a},0,0)} & \cdots & T_{(0,0),(C,D)}\mathbf{E}_{\delta(\mathbf{a},0,0)} \\ \vdots & \ddots & \vdots \\ T_{(C,D),(0,0)}\mathbf{E}_{\delta(\mathbf{a},C,D)} & \cdots & T_{(C,D),(C,D)}\mathbf{E}_{\delta(\mathbf{a},C,D)} \end{bmatrix}. \quad (5)$$

Here the net change of the energy level is defined as a function of the action $\mathbf{a}$, the current channel state c and the current traffic state d. The function is expressed as follows:

$$\delta\big(\mathbf{a}=(a_1,a_2),c,d\big) = \begin{cases} 0, & a_1 = 0 \text{ and } a_2 = 0, \\ c, & a_1 = 1 \text{ and } a_2 = 0, \\ c-1, & a_1 = 1 \text{ and } a_2 = 1, \\ -1, & a_1 = 0 \text{ and } a_2 = 1. \end{cases} \quad (6)$$

The first case is for when the user does not request for energy and does not transmit data. The second case is for when the user requests for energy and does not transmit (i.e., the energy level may increase according to the channel state). The third case is for when the user requests for energy and transmits data (i.e., the energy level may increase by c (according to the channel state) minus one unit for data transmission). The fourth case is for when the user does not request for energy and transmits data (i.e., the energy level decreases by one unit).

Next, we define the immediate reward $\mathscr{R}_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a})$ based on the user's composite state $\omega_{\mathrm{U}} = (e, c, d)$ and composite action $\mathbf{a}$ as follows:

$$\mathscr{R}_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) = \begin{cases} u_d^{\mathrm{U}}\big(1-\rho(\pi_{\mathrm{A}})\big), & a_1 = 0 \text{ and } a_2 = 1 \\ & \text{ and } e > 0, \\ u_d^{\mathrm{U}}\big(1-\rho(\pi_{\mathrm{A}})\big) - \mu, & a_1 = 1 \text{ and } a_2 = 1 \\ & \text{ and } e > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

$\rho(\pi_{\mathrm{A}})$ is the attack probability which is defined as a function of an attack policy π_{A} of the attacker (to be given in (34)). The first case is for when the user does not request for energy. The immediate reward is the utility of successfully transmitting the data generated at traffic state d (i.e., there is no jamming attack). The second case is for when the user requests for energy, and hence, the user has to pay the cost of μ.

We define the energy cost as a constraint as follows:

$$\mathscr{V}_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) = \begin{cases} \mu, & a_1 = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Let π_{U} denote a stationary policy of the user. In particular, $\pi_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a})$ is the probability that the user takes a local action $\mathbf{a}$ at local state ω_{U}. The user aims to maximize a long-term average reward, which is defined as follows:

$$R_{\mathrm{U}}(\pi_{\mathrm{U}}, \pi_{\mathrm{A}}) = \lim_{t'\to\infty} \frac{1}{t'} \sum_{t=1}^{t'} E_{\pi_{\mathrm{U}}}\Big(\mathscr{R}_{\mathrm{U}}(\omega_{\mathrm{U}}(t), \mathbf{a}(t))\Big), \quad (9)$$

where $\omega_{\mathrm{U}}(t)$ and $\mathbf{a}(t)$ are the local state and global action at time t, respectively. Moreover, the user must meet the constraint on the long-term average energy cost, defined as follows:

$$V_{\mathrm{U}}(\pi_{\mathrm{U}}, \pi_{\mathrm{A}}) = \lim_{t'\to\infty} \frac{1}{t'} \sum_{t=1}^{t'} E_{\pi_{\mathrm{U}}}\Big(\mathscr{V}_{\mathrm{U}}(\omega_{\mathrm{U}}(t), \mathbf{a}(t))\Big) \leq K, \quad (10)$$

where K is the energy cost threshold of the user.

The best response policy π_{U}^* of the user given any policy of the attacker π_{A} is defined as follows:

$$R_{\mathrm{U}}\Big((\pi_{\mathrm{U}}^*, \pi_{\mathrm{A}})\Big) \geq R_{\mathrm{U}}\Big((\pi_{\mathrm{U}}, \pi_{\mathrm{A}})\Big). \quad (11)$$

We apply a linear programming approach to obtain the best response policy of the user. Let $\phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a})$ denote the stationary probability of a local state ω_{U} and local action $\mathbf{a}$ of the user given policy of the attacker π_{A}. We solve the following optimization to obtain $\phi_{\pi_{\mathrm{A}}}^{\mathrm{U}*}(\omega_{\mathrm{U}}, \mathbf{a})$,

$$\max_{\phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}},\mathbf{a})} \quad \sum_{\omega_{\mathrm{U}}\in\Omega_{\mathrm{U}}} \sum_{\mathbf{a}\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) \mathscr{R}_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) \quad (12)$$

$$\text{s.t.} \quad \sum_{\omega_{\mathrm{U}}\in\Omega_{\mathrm{U}}} \sum_{\mathbf{a}\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) \mathscr{V}_{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) \leq K \quad (13)$$

$$\sum_{\omega_{\mathrm{U}}\in\Omega_{\mathrm{U}}} \sum_{\mathbf{a}\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) P_{\omega_{\mathrm{U}},\omega'_{\mathrm{U}}}(\mathbf{a}) = \sum_{\mathbf{a}\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega'_{\mathrm{U}}, \mathbf{a}), \omega'_{\mathrm{U}} \in \Omega_{\mathrm{U}} \quad (14)$$

$$\sum_{\omega_{\mathrm{U}}\in\Omega_{\mathrm{U}}} \sum_{\mathbf{a}\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) = 1, \quad (15)$$

$$\phi_{\pi_{\mathrm{A}}}^{\mathrm{U}}(\omega_{\mathrm{U}}, \mathbf{a}) \geq 0, \quad (16)$$

where $P_{\omega_{\mathrm{U}},\omega'_{\mathrm{U}}}(\mathbf{a})$ denotes the element of matrix $\mathbf{P}_{\mathrm{U}}(\mathbf{a})$ as defined in (5). Assume that the linear programming problem is feasible, the stationary best response policy of the user can be obtained as follows:

$$\pi_{\mathrm{U}}^*(\omega_{\mathrm{U}}, \mathbf{a}) = \frac{\phi_{\pi_{\mathrm{A}}}^{\mathrm{U}*}(\omega_{\mathrm{U}}, \mathbf{a})}{\sum_{\mathbf{a}'\in\mathcal{A}_{\mathrm{U}}} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}*}(\omega_{\mathrm{U}}, \mathbf{a}')}, \text{ for } \omega_{\mathrm{U}} \in \Omega_{\mathrm{U}}. \quad (17)$$

Given the best response policy of the user, the energy request probability is obtained from

$$\lambda(\pi_{\mathrm{U}}) = \sum_{\omega_{\mathrm{U}}\in\Omega_{\mathrm{U}}} \sum_{a_1=1} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}*}(\omega_{\mathrm{U}}, \mathbf{a}). \quad (18)$$

The data transmission probability is obtained from

$$\tau(\pi_{\mathrm{U}}) = \sum_{e=1}^{E_{\mathrm{U}}} \sum_{c=0}^{C} \sum_{d=1}^{D} \sum_{a_2=1} \phi_{\pi_{\mathrm{A}}}^{\mathrm{U}*}(\omega_{\mathrm{U}}, \mathbf{a}). \quad (19)$$

The energy request and data transmission probabilities of the user will affect the policy of the attacker to jam the data transmission (e.g., as in (22) and (26)).

B. Attacker Side Analysis

The state space of the attacker is defined as follows:

$$\Omega_{\mathrm{A}} = \Big\{ (\mathbb{E}, \mathbb{C}); \mathbb{E} \in \{0, 1, \ldots, E_{\mathrm{A}}\} \Big\}, \tag{20}$$

where $\mathbb{E}$ is the energy level of the energy storage and $\mathbb{C} \in \mathcal{C}_{\mathrm{A}}$ is the channel state. The composite state variable of the attacker is denoted by $\omega_{\mathrm{A}} = (e, c)$.

The action space is defined as $\mathcal{A}_{\mathrm{A}} = \{0, 1\}$, where 0 and 1 represent the actions of not attack and attack, respectively. $a \in \mathcal{A}_{\mathrm{A}}$ is an action variable.

Next, we derive the transition matrix for the attacker. Again, we first consider the transition of energy level. The transition matrix is denoted by $\mathbf{B}_{\delta,a}$, where $\delta = 0, 1, \ldots, C$ is the amount of incoming energy. For action $a = 0$ (i.e., no attack), the element of the matrix $\mathbf{B}_{\delta,a}$, denoted by $B^{\delta,a}_{b,b'}$ (i.e., the probability that the energy level changes from b to b'), is obtained from

$$B^{\delta=0,a=0}_{b,b'} = \begin{cases} 1, & b = b', \\ 0, & \text{otherwise}, \end{cases} \tag{21}$$

$$B^{\delta>0,a=0}_{b,b'} = \begin{cases} \lambda(\pi_{\mathrm{U}}), & b' = \min(E_{\mathrm{A}}, b + \delta), \\ 1 - \lambda(\pi_{\mathrm{U}}), & b' = b, \\ 1, & b' = b = E_{\mathrm{A}}, \\ 0, & \text{otherwise}, \end{cases} \tag{22}$$

where $\lambda(\pi_{\mathrm{U}})$ is the probability that the wireless energy source transfers energy (i.e., the user requests for energy transfer). This probability is a function of an energy request and data transmission policy π_{U} of the user. For action $a = 1$ (i.e., attack), the element of the matrix $\mathbf{B}_{\delta,a}$ is obtained from

$$B^{\delta=0,a=1}_{b,b'} = \begin{cases} 1, & b' = \max(0, b - 1), \\ 0, & \text{otherwise}, \end{cases} \tag{23}$$

$$B^{\delta>0,a=1}_{b,b'} = \begin{cases} \lambda(\pi_{\mathrm{U}}), & b' = \min(E_{\mathrm{A}}, b + \delta), \\ 1 - \lambda(\pi_{\mathrm{U}}), & b' = \max(0, b - 1), \\ 0, & \text{otherwise}. \end{cases} \tag{24}$$

Then, the transition matrix of the attacker, denoted by $\mathbf{P}_{\mathrm{A}}(a)$, is obtained from

$$\mathbf{P}_{\mathrm{A}}(a) = \begin{bmatrix} C^{\mathrm{A}}_{0,0}\mathbf{B}_{\delta=0,a} & \cdots & C^{\mathrm{A}}_{0,C}\mathbf{B}_{\delta=0,a} \\ \vdots & \ddots & \vdots \\ C^{\mathrm{A}}_{C,0}\mathbf{B}_{\delta=C,a} & \cdots & C^{\mathrm{A}}_{C,C}\mathbf{B}_{\delta=C,a} \end{bmatrix} \tag{25}$$

where $C^{\mathrm{A}}_{c,c'}$ is the element of transition matrix $\mathbf{C}_{\mathrm{A}}$ of the channel state of the attacker.

We define the immediate reward $\mathscr{R}_{\mathrm{A}}(\omega_{\mathrm{A}}, a)$ based on the attacker's composite state $\omega_{\mathrm{A}} = (e, c)$ and action a as follows:

$$\mathscr{R}_{\mathrm{A}}(\omega_{\mathrm{A}}, a) = \begin{cases} 1 - \tau(\pi_{\mathrm{U}}), & a = 1 \text{ and } b > 0, \\ 0, & \text{otherwise}. \end{cases} \tag{26}$$

$\tau(\pi_{\mathrm{U}})$ is the data transmission probability which is defined as a function of the policy π_{U} of the user as in (19).

Let π_{A} denote a stationary policy of the attacker. In particular, $\pi_{\mathrm{A}}(\omega_{\mathrm{A}}, a)$ is the probability that the attacker takes a local action a at a local state ω_{A}. The attacker aims to maximize a long-term average reward, which is defined as follows:

$$R_{\mathrm{A}}(\pi_{\mathrm{A}}, \pi_{\mathrm{U}}) = \lim_{t' \to \infty} \frac{1}{t'} \sum_{t=1}^{t'} E_{\pi_{\mathrm{A}}} \Big(\mathscr{R}_{\mathrm{A}}(\omega_{\mathrm{A}}(t), a(t)) \Big), \tag{27}$$

where $\omega_{\mathrm{A}}(t)$ and $a(t)$ are the local state and global action at time t, respectively.

Similarly, the best response policy π^*_{A} of the attacker given any policy of the user π_{U} is defined as follows:

$$R_{\mathrm{A}}\Big((\pi^*_{\mathrm{A}}, \pi_{\mathrm{U}})\Big) \geq R_{\mathrm{A}}\Big((\pi_{\mathrm{A}}, \pi_{\mathrm{U}})\Big). \tag{28}$$

Using the linear programming approach, let $\phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a)$ denote the stationary probability of a local state ω_{A} and local action a of the attacker given policy of the user π_{U}. We solve the following optimization to obtain $\phi^{\mathrm{A}*}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a)$,

$$\max_{\phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a)} \quad \sum_{\omega_{\mathrm{A}} \in \Omega_{\mathrm{A}}} \sum_{a \in \mathcal{A}_{\mathrm{A}}} \phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a) \mathscr{R}_{\mathrm{A}}(\omega_{\mathrm{A}}, a) \tag{29}$$

$$\text{s.t.} \quad \sum_{\omega_{\mathrm{A}} \in \Omega_{\mathrm{A}}} \sum_{a \in \mathcal{A}_{\mathrm{A}}} \phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a) P_{\omega_{\mathrm{A}}, \omega'_{\mathrm{A}}}(a) = \sum_{a \in \mathcal{A}_{\mathrm{A}}} \phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega'_{\mathrm{A}}, a), \omega'_{\mathrm{A}} \in \Omega_{\mathrm{A}}, \tag{30}$$

$$\sum_{\omega_{\mathrm{A}} \in \Omega_{\mathrm{A}}} \sum_{a \in \mathcal{A}_{\mathrm{A}}} \phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a) = 1, \tag{31}$$

$$\phi^{\mathrm{A}}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a) \geq 0, \tag{32}$$

where $P_{\omega_{\mathrm{A}}, \omega'_{\mathrm{A}}}(a)$ denotes the element of matrix $\mathbf{P}_{\mathrm{A}}(a)$ as defined in (25). The stationary best response policy of the attacker can be obtained as follows:

$$\pi^*_{\mathrm{A}}(\omega_{\mathrm{A}}, a) = \frac{\phi^{\mathrm{A}*}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a)}{\sum_{a' \in \mathcal{A}_{\mathrm{A}}} \phi^{\mathrm{A}*}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}}, a')}, \text{ for } \omega_{\mathrm{A}} \in \Omega_{\mathrm{A}}. \tag{33}$$

Given the best response policy of the attacker, the attack probability is obtained from

$$\rho(\pi_{\mathrm{A}}) = \sum_{e=1}^{E_{\mathrm{A}}} \sum_{c=0}^{C} \sum_{a=1} \phi^{\mathrm{A}*}_{\pi_{\mathrm{U}}}(\omega_{\mathrm{A}} = (e, c), a). \tag{34}$$

This attack probability of the attacker will affect the policy of the user (i.e., in (7)).

C. Solution and Algorithm

We consider a constrained Nash equilibrium as the solution of the constrained stochastic game between the user and attacker. The policy π^*_{U} and π^*_{A} are the constrained Nash equilibrium for the user and attacker, respectively, if

$$R_{\mathrm{U}}\Big((\pi^*_{\mathrm{U}}, \pi^*_{\mathrm{A}})\Big) \geq R_{\mathrm{U}}\Big((\pi_{\mathrm{U}}, \pi^*_{\mathrm{A}})\Big), \tag{35}$$

$$R_{\mathrm{A}}\Big((\pi^*_{\mathrm{A}}, \pi^*_{\mathrm{U}})\Big) \geq R_{\mathrm{A}}\Big((\pi_{\mathrm{A}}, \pi^*_{\mathrm{U}})\Big), \tag{36}$$

where $R_{\mathrm{U}}\Big((\pi_{\mathrm{U}}, \pi^*_{\mathrm{A}})\Big)$ and $R_{\mathrm{A}}\Big((\pi_{\mathrm{A}}, \pi^*_{\mathrm{U}})\Big)$ are feasible (i.e., the constraint is met). The constrained Nash equilibrium ensures that the player cannot achieve higher reward by adopting any other stationary policies while the other player does not change their stationary policy.

We apply best response dynamics to obtain the constrained Nash equilibrium as shown in Algorithm 1.

Algorithm 1 Best response dynamics

1: Initialize the policies π_{U} and π_{A} for the user and attacker
2: **repeat**
3: Calculate the energy request and data transmission probabilities of the user (i.e., $\lambda(\pi_{\mathrm{U}})$ and $\tau(\pi_{\mathrm{U}})$ from (18) and (19), respectively) given the policy π_{A} of the attacker
4: Attacker obtains its best response policy π_{A} given the energy request and data transmission probabilities of the user
5: Calculate the attack probability of the attacker (i.e., $\rho(\pi_{\mathrm{A}})$ from (34) given the policy π_{U} of the user
6: User obtains its best response policy π_{U} given the attack probability
7: **until** Stationary policies of the user and attacker converge

IV. Performance Evaluation

A. Parameter Setting

We consider a wireless powered network with an attacker as shown in Fig. 1. A user's device has an energy storage with the capacity of 20 units of energy. There are four channel states, and the probabilities of channel states 0, 1, 2, and 3 are 0.4, 0.3, 0.2, and 0.1, respectively. There are three traffic states, and the probabilities to be in traffic states 1, 2, and 3 are 0.4, 0.3, and 0.3, respectively. The utility of successfully transmitting data generated from traffic states 1, 2, and 3, are 1, 2, and 3, respectively, for the user. The cost of requesting energy for the user is one. Unless otherwise stated, the energy cost threshold is set to one. The attacker has the energy storage with the same size and with the same channel state probabilities. However, the channel states of the user and the attacker are independent. The utility of successfully jamming the data transmission of the user is one for the attacker. For the iterative algorithm, we set initial values of energy request probability, data transmission probability, and jamming attack probability at 0.5.

B. Numerical Results

We first show the equilibrium policies of the user and attacker in Fig. 2. Note that "(Traffic,Channel)" denotes a composite state of traffic state and channel state. Figure 2(a) shows that the user will request for wireless energy transfer when the channel state is good (i.e., $c = 2, 3$). If channel state is $c = 1$, the user will request for wireless energy only when the energy level in the energy storage is low, and the user will never request when the channel state is $c = 0$. This policy is expected as the user has to pay for wireless energy transfer. As a result, the user has to maximize the amount energy received by selectively requesting for wireless energy when the channel is good. Figure 2(b) shows the policy of the user to transmit data. Similarly, the user will transmit data if the traffic state is high. If the traffic state is low, the user selectively transmits data only when the channel state is good and there are more amount of energy in the storage. This is to ensure that when the traffic state is high, there will be enough energy to transmit the data so that the user gains the maximum utility. Figure 2(c) shows the policy of the attacker. There is no clear trend of the attacker to jam the user, as this is a randomized policy. However, it is likely that when there are more energy in the storage, the attacker will jam the transmission of the user.

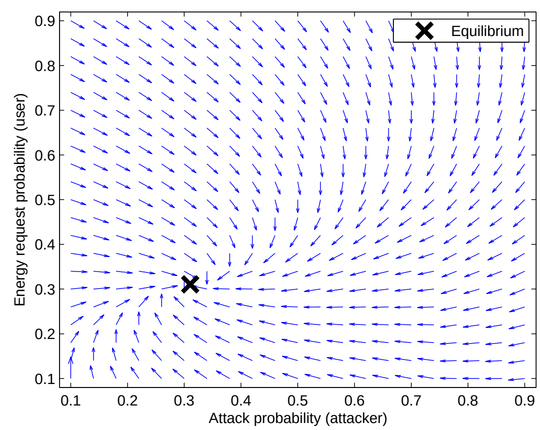
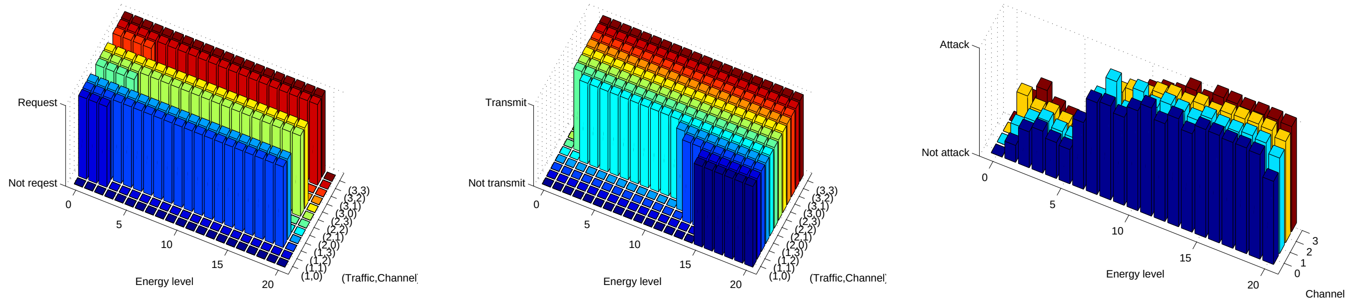


Fig. 3. Convergence toward equilibrium solution of user and attacker.

Figure 3 shows the attraction of the energy request probability of the user and the attack probability of the attacker to the equilibrium. We observe that at a wide range of the initial probabilities, the iterative algorithm will converge to the equilibrium solution. Figure 4 shows the energy request probability, data transmission probability, and attack probability of the user and attacker. As the energy cost per request increases, the user will request less frequently for wireless energy transfer. As a result, there is less energy supply and the user can transmit less data. Similarly, the attacker can receive less energy, and it will perform less jamming attack to the user. Figure 4 also shows the simulation results which match well with the analytical results obtained from the optimization.

Figures 5(a) and (b) show the attack probability of the attacker and average reward of the user, respectively. When the energy cost threshold increases, the user will request more energy, and hence, the attacker will receive more wireless energy. Consequently, the attacker has more resource to perform the jamming attack. However, at a certain poind, since requesting wireless energy also incurs cost, the user will not increase the energy request probability, although the threshold still keeps increasing. Therefore, the attack probability remains constant at a certain point. Here, we observe an interesting result that the average reward of the user first increases, then decreases and remains constant as the energy cost threshold increases.



(a) (b) (c)

Fig. 2. (a) Energy request policy of user, (b) data transmission policy of user, and (c) jamming attack policy of attacker.

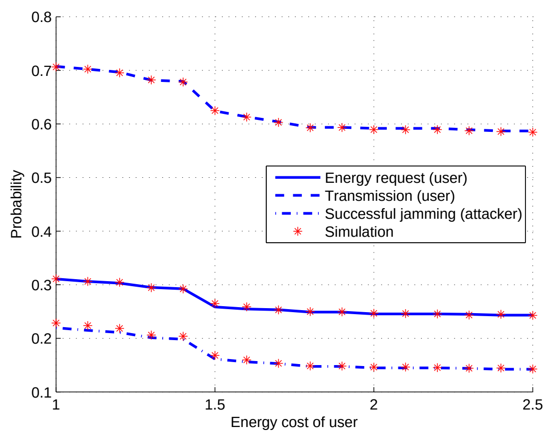
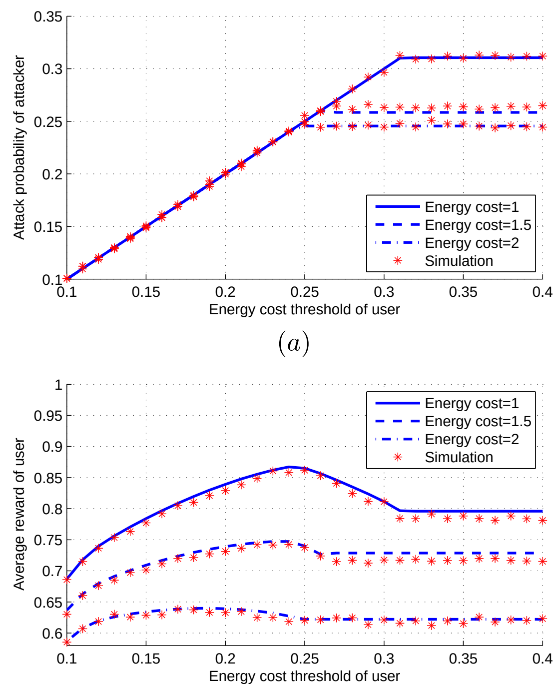


Fig. 4. Probabilities of energy request by user, data transmission by user, and jamming attack by attacker.

The average reward decreases since the user requests too much wireless energy, incurring too much cost and experiencing much attacks. Therefore, at a certain point, the user stops requesting more energy and the average reward becomes a constant.



(b)

Fig. 5. (a) Jamming attack probability of attacker and (b) average reward of user.

V. Summary

Jamming attack can happen in wireless powered networks, where an attacker receives and uses energy from a wireless energy source to attack a legitimate user. The jamming attack results in failed data transmission of the user. In this paper, we have proposed a game theoretic analysis of the jamming attack, where the user and attacker are aware of each other. They optimize their policies (i.e., the energy request and data transmission policy for the user and the attack policy of the attacker). We have proposed an iterative algorithm to analyze the proposed equilibrium policies. The performance evaluation has shown the convergence of the algorithm. Additionally, we have found that there is an energy cost threshold that can maximize the user's average reward.

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